SMAI ASSIGNMENT 2

Name: Shreya Koka
Roll No: 2024202004

Part 1: Face Recognition

1. Accuracy, Precision, Recall, F1-Scores

VGGFace (Finetuning)

- Accuracy: 82.5%
- Precision: 0.71
- Recall: 1.00
- F1 Score: 0.83

ResNet18 (from scratch)

- Accuracy: 82.5%
- Precision: 0.7083
- Recall: 1.0000
- F1 Score: 0.8293

ResNet18 (from pretrained on ImageNet)

- Accuracy: 60%
- Precision: 0.5385
- Recall: 0.4118
- F1 Score: 0.4667

2. Qualitative Analysis of the models

VGGFace (Finetuning)

- Excels at identifying your_face with perfect recall, leveraging FaceNet's pretrained weights optimized for facial features. Moderate precision indicates false positives (e.g., other faces misclassified as yours), likely due to variations in lighting, angles, or backgrounds in the test set of final.zip (~40 images).

ResNet18 (from scratch)

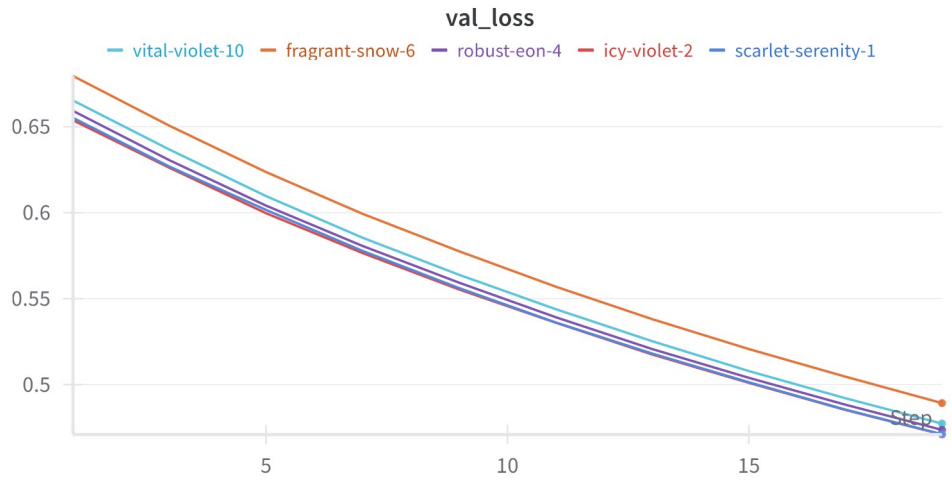
- Matches VGGFace's performance despite no pretraining, relying on the quality of final.zip's train folder. High recall ensures all positive cases are captured, but moderate precision suggests false positives in varied test conditions. Effective when training and test images share similar characteristics (e.g., same subjects, scenarios).

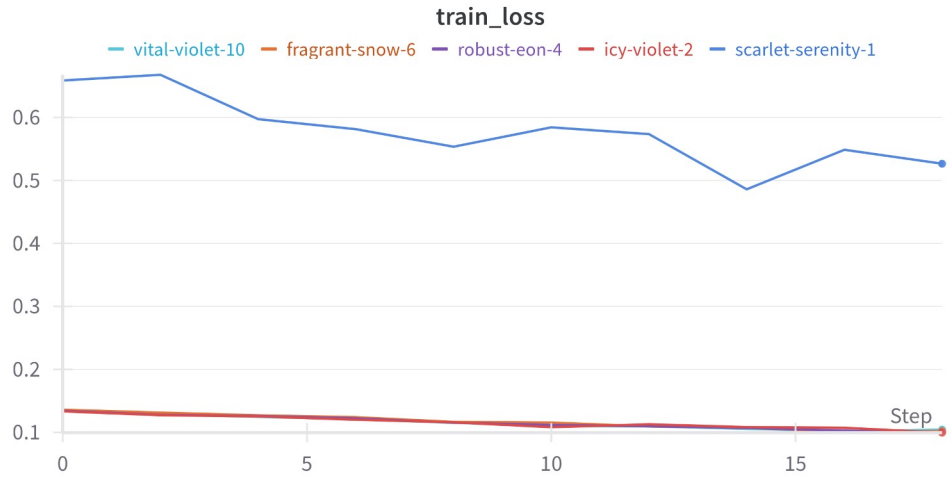
ResNet18 (from pretrained on ImageNet)

- ImageNet pretraining, suited for general objects, struggles with face-specific features, resulting in low recall (missing many instances of your face) and precision (high false positives/negatives). Poor performance in distinguishing fine-grained facial differences, exacerbated by the potentially small or diverse test set in final.zip. Weak across most conditions, particularly with small or varied test sets; ImageNet weights are ill-suited for face recognition.

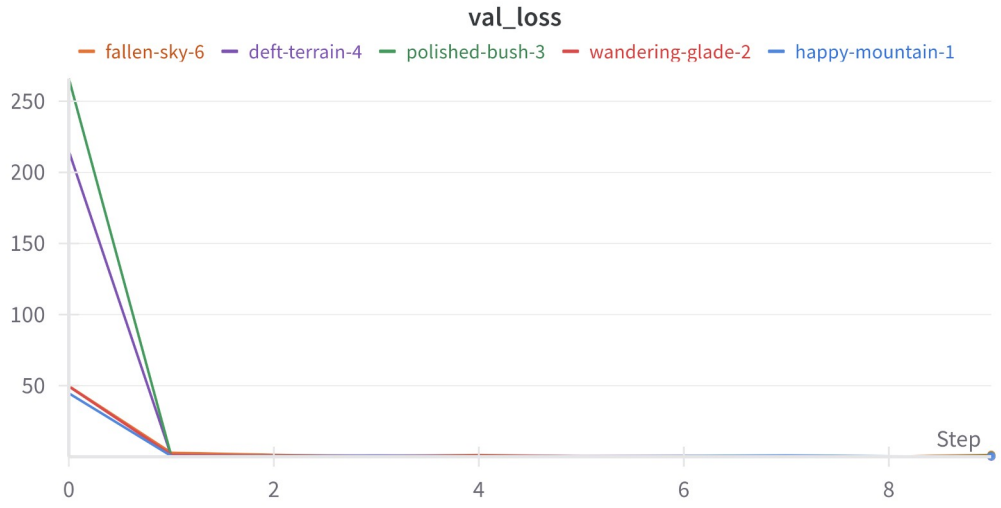
3. Train and validation curves

VGGFace (Finetuning)





ResNet18 (from scratch)





ResNet18 (from pretrained on ImageNet)





4. Challenges faced while training on custom dataset

- Learning Rate Tuning (VGGFace): Fine-tuning FaceNet weights required a low learning rate (e.g., 0.0001) to avoid overfitting, as large updates could disrupt pretrained facial feature representations.
- Test Set Variability (All Models): Variations in the test folder (e.g., lighting, angles, backgrounds) caused false positives, challenging model generalization, especially for ResNet18-Pretrained, which lacked face-specific features.
- Feature Misalignment (ResNet18-Pretrained): ImageNet weights were poorly suited for faces, leading to slow adaptation and low performance, exacerbated by the potentially small train folder size.
- Colab Resource Limits (All Models): Training in Colab risked session crashes with large datasets, requiring optimized batch sizes (e.g., 16) and limited NUM_WORKERS (e.g., 2 for GPU, 0 for CPU).

Part 2: Emotion Recognition

1. Accuracy per emotion class

VGGFace (Finetuning)

- `Test Per-class Accuracies: {'angry': 0.7142857142857143, 'happy': 0.7777777777777778, 'sad': 0.7954545454545454}`

ResNet18 (from scratch)

- `Test Per-class Accuracies: {'angry': 0.7142857142857143, 'happy': 0.4444444444444444, 'sad': 0.9318181818181818}`

ResNet18 (from pretrained on ImageNet)

- `Test Per-class Accuracies: {'angry': 1.0, 'happy': 0.9777777777777777, 'sad': 0.8409090909090909}`

2. Qualitative Analysis of the models

VGGFace (Finetuning)

- Balanced performance across emotions, with sad recognition strongest (79.55%). FaceNet's face-specific pretraining aids in capturing facial expressions, but moderate accuracies suggest challenges with overlapping features (e.g., angry vs. sad) in test_raw.zip.

ResNet18 (from scratch)

- Strong sad recognition (93.18%) but very poor happy recognition (44.44%), leading to low overall accuracy. Without pretraining, it relies on raw.zip's quality, excelling for distinct emotions (sad) but failing for complex or underrepresented ones (happy), likely due to limited training data or imbalance. Best with emotions having consistent features in training; weak with diverse or underrepresented expressions (e.g., happy's varied smiles).

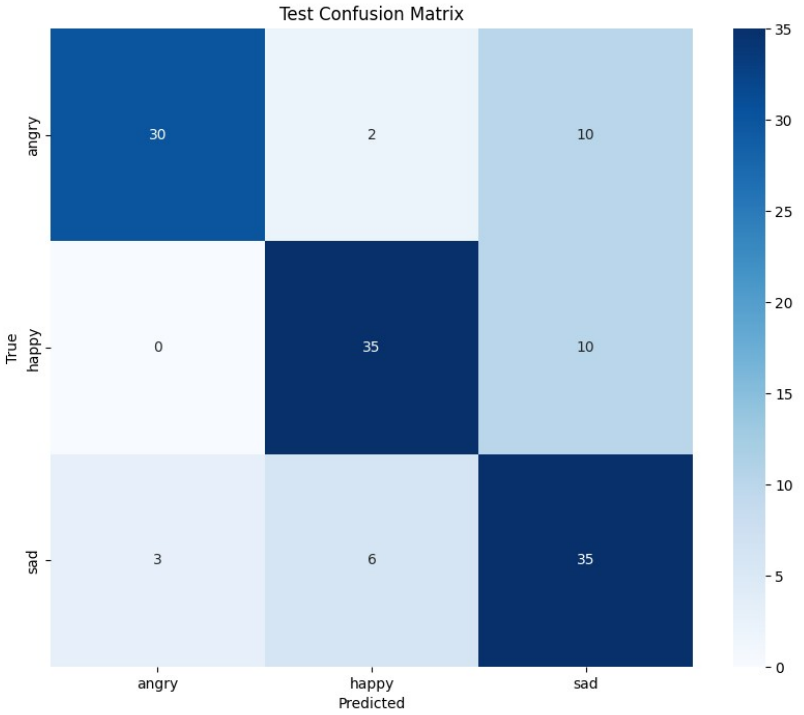
ResNet18 (from pretrained on ImageNet)

- Outperforms others, with perfect angry recognition and near-perfect happy recognition. ImageNet pretraining, fine-tuned on raw.zip, captures general visual patterns effectively, though slightly lower sad accuracy (84.09%) suggests difficulty with subtler expressions. Robust across emotions due to pretrained feature diversity. Adapts well to diverse test conditions (e.g., varied lighting, angles).

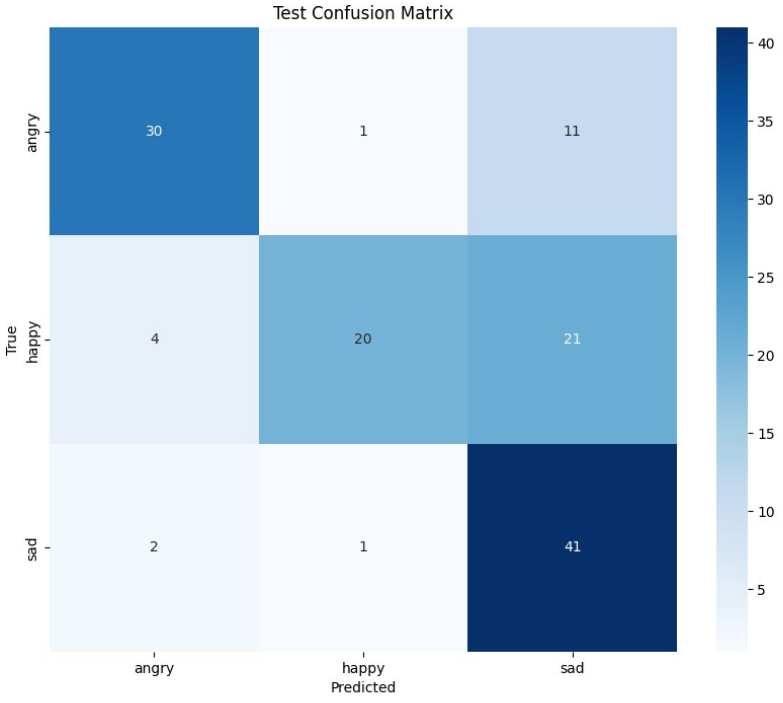
Variability (e.g., angles, lighting) challenged ResNet18-Scratch most, while pretrained models handled diversity better.

3. Confusion matrix

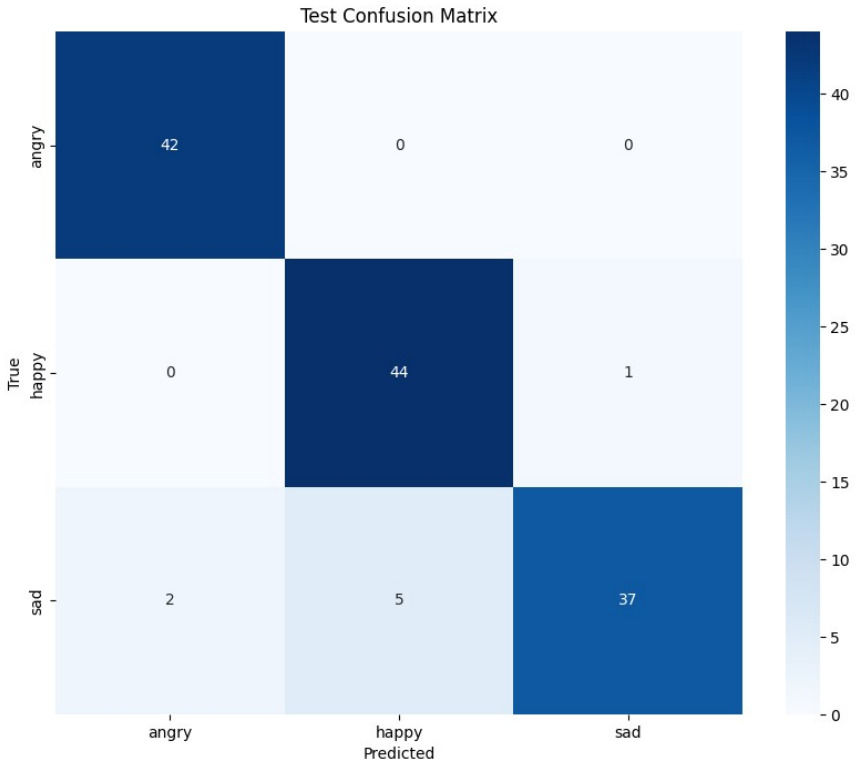
VGGFace (Finetuning)



ResNet18 (from scratch)

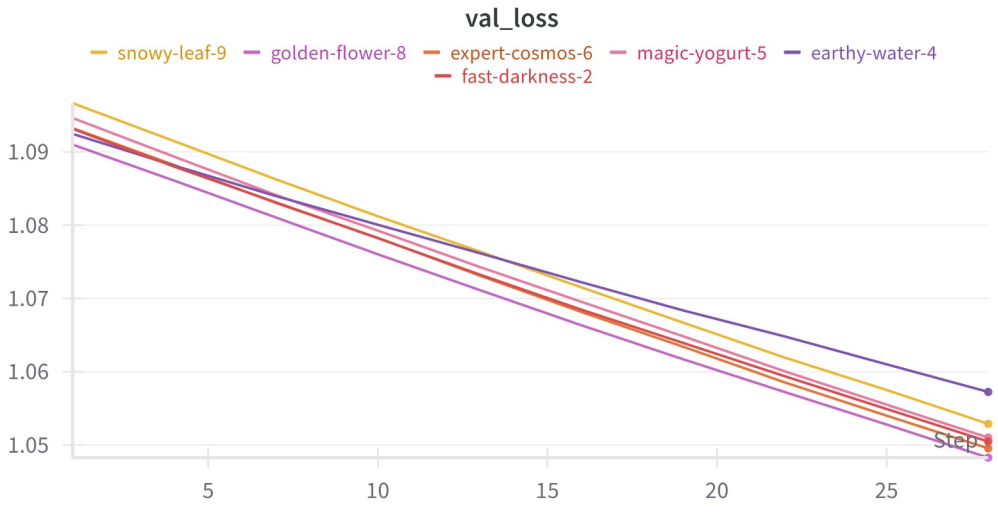


ResNet18 (from pretrained on ImageNet)



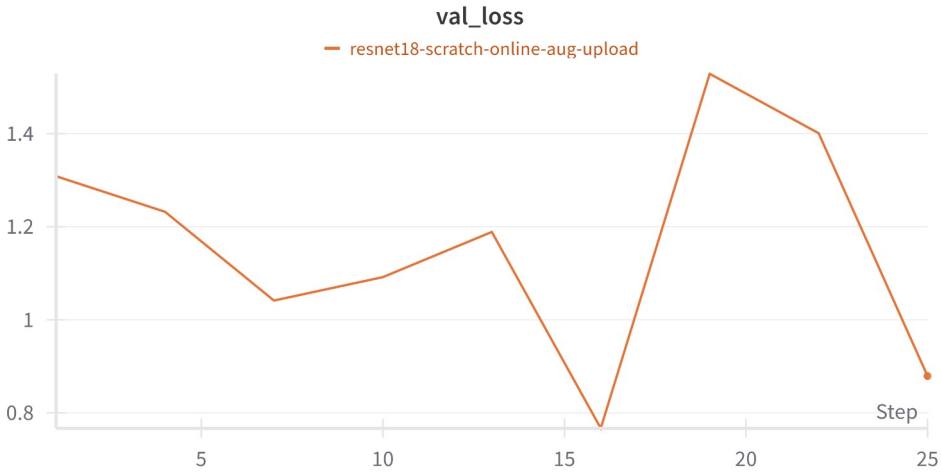
4. Training curves (loss vs epoch, accuracy vs epoch)

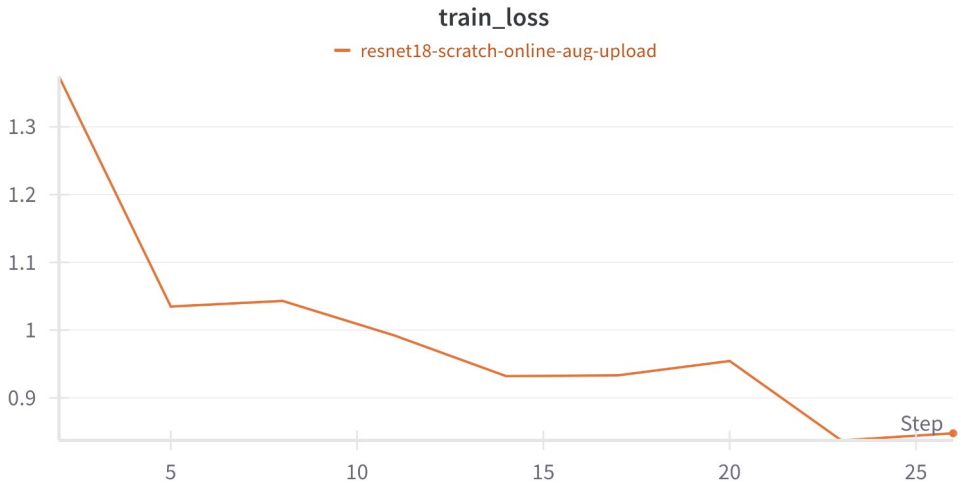
VGGFace (Finetuning)



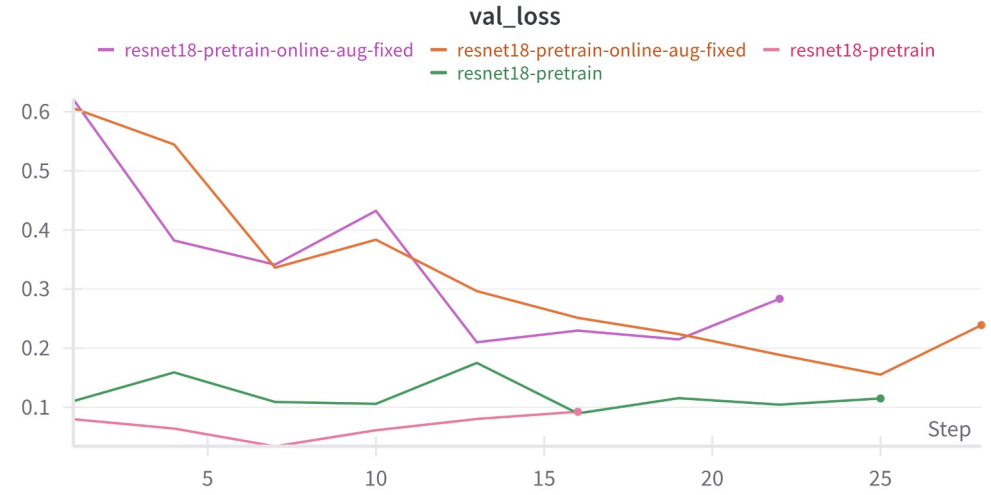


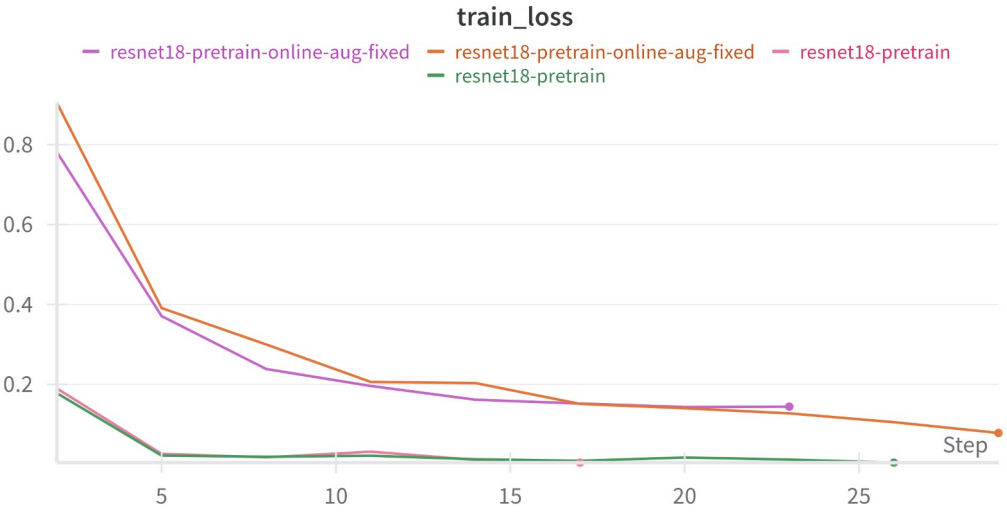
ResNet18 (from scratch)





ResNet18 (from pretrained on ImageNet)





5. <u>Challenges faced while training on custom dataset</u>

- Class Imbalance (All Models): Uneven emotion distribution in training data (e.g., fewer happy images) led to poor performance, especially for ResNet18-Scratch on happy (44.44%). Stratified splitting and augmentation (e.g., flips, rotations) were needed to mitigate bias.
- Subtle Emotion Differences (All Models): Overlapping features (e.g., angry vs. sad) in testing data caused misclassifications, particularly for VGGFace and ResNet18-Scratch, requiring careful augmentation to preserve expression details.
- Slow Convergence (ResNet18-Scratch): Training from scratch demanded more epochs (50) and robust augmentation to learn emotion-specific features, increasing computational cost and yielding uneven performance (low happy accuracy).

- Feature Adaptation (ResNet18-Pretrained): ImageNet weights required significant fine-tuning to adapt to facial expressions, though training dataset's structure enabled strong performance after optimization.
- Colab Resource Limits (All Models): Training in Colab risked session crashes with large datasets, requiring optimized batch sizes (e.g., 16) and limited NUM_WORKERS (e.g., 2 for GPU, 0 for CPU).